A Simple Proof for the Hamiltonian Property on Rotation Graphs of Binary Trees

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Abstract

In [J.M. Lucas, The rotation graph of binary trees is Hamiltonian, Journal of Algorithms 8 (1987)], Lucas proved that the rotation graph of binary trees contains a Hamiltonian cycle by using the codewords representation of binary trees. In this paper, we use the weight sequences introduced in [J.M. Pallo, Enumerating, ranking and unranking binary trees, Computer Journal 29 (1986)] to prove the existence of Hamiltonian cycles in rotation graphs. With the assistance of recursion trees and the Gray-code order of weight sequences, a conceptually easy proof is given.

Keywords: binary trees; rotation graphs; Hamiltonian cycles

1 Introduction

Binary search trees are elementary data structures which stores data with prescribed order. When constructing a binary search tree online without any modification, the resulting tree may be skew. The worst case search time depends on how skew the binary search tree is. To avoid this situation, researchers pay attention to balancing the resulting binary search tree. The rotation operation is one of the general technique used for this purpose. Figure 1 shows how rotation is performed in a binary tree. According to the rotation operation, the *rotation graph* of binary trees is defined. A rotation graph RG(n) is the graph with vertex set being *ordered* binary trees with n internal vertices, where

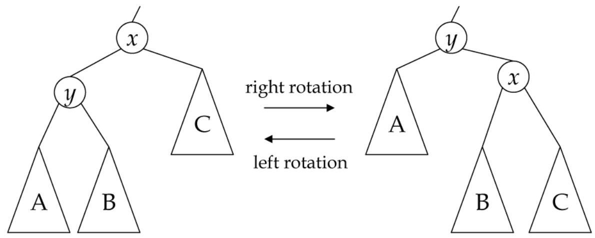

Figure 1: Rotation on edge (x, y).

a tree is ordered if the relative order of the subtrees is fixed. Two trees are adjacent if one tree can be transformed to the other by a single rotation. Properties, like the Hamiltonian path, the Hamiltonian cycle, and the rotation distance, in RG(n) are widely explored [1]. Generating binary trees is also an important topic in this line of investigation. Most generating algorithms use an integer sequence to represent a binary tree. Some algorithms generate the integer sequences such that two continuous sequences differ in constant positions, which implies the structural similarity between two trees, and a Hamiltonian path in the corresponding rotation graph is derived.

Lucas [1] showed that rotation graphs contain Hamiltonian cycles by using the codeword to represent a binary tree. The equivalence between the codeword and that proposed by Zerling [12] was shown, and a bijection to binary trees is derived. Lucas defined a general structure, the *stack*, to show the existence of a Hamiltonian path between two adjacent vertices in a rotation graph. The proof is a complicated induction.

Pallo [6] introduced the *weight sequence* representation of binary trees. Roelants van Baronaigien and Ruskey [4] proposed an algorithm to recursively generate the weight sequence in Gray-code order, i.e., two continuous sequences different in one position. The same result was also achieved by Vajnovszki [9] using a loopless algorithm. In the ordered set of sequences generated by the algorithm in [4, 9], two continuous weight sequences are adjacent in the rotation graph. As a consequence, a Hamiltonian path in the rotation graph is derived. However, the end vertices are not adjacent. Our strategy for constructing a Hamiltonian cycle is to insert and to substitute some edges in that path.

The rest of the paper is organized as follows. In Section 2, we summarize some properties about weight sequences and the recursion trees when generating the sequences. In Section 3, we construct a Hamiltonian cycle in RG(n). Finally, concluding remarks are given in Section 4.

2 Properties on Weight Sequences

In this section, we define the notation and summarize some properties about the weight sequence. Readers can refer to Harary [5] for any graph-theory terms not defined here. In the rest of this paper, for simplicity, a tree refers to an ordered binary tree, *vertices* and *nodes* refer to the entries in the vertex set of a binary tree and a recursion tree, respectively, and vertex i is the ith one when traversing the internal vertices of a given binary tree in inorder. The *size* of a tree T, which is denoted by $|T|$, is the number of internal vertices in T. The tree T_x denotes the subtree of T induced by all descendent of vertex x. A cycle of n nodes is denoted by C_n, for $n \geq 1$.

Given a tree T of size n, the weight sequence of T is the sequence $s_1 s_2 \cdots s_n$, where s_i is the number of leaves in the left subtree of vertex i. An example is shown in Figure 2 There is a bijection

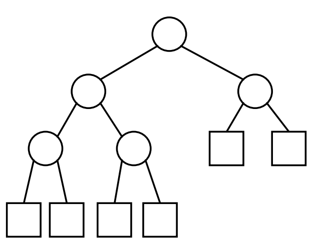

Figure 2: The tree with weight sequence 12141, where internal nodes and leaves are denoted by circles and rectangles, respectively.

between weight sequences of length n and trees of size n. We denote the tree corresponding to weight sequence s by $T(s)$. In a tree T of size n, the *right arm* of T is the path from the root to vertex n. Given a weight sequence $s_1 s_2 \cdots s_n$, we can construct $T(s_1 s_2 \cdots s_n)$ by inserting vertex n to the tree $T(s_1 s_2 \cdots s_{n-1})$. Let $P_r = (x_1, x_2, \ldots, x_k)$ be the right arm of $T(s_1 s_2 \cdots s_{n-1})$, where x_i is the parent of x_{i+1} for $1 \leq i < k$ and x_1 is the root. If $s_n = 1$ or n, the tree $T(s_1 s_2 \cdots s_n)$ is constructed by making vertex n the right child of x_k or the right parent of x_1, respectively. Otherwise, $T(s_1 s_2 \cdots s_n)$ is constructed by making node n the right child of x_i and the right parent of x_{i+1}, where $|T_{x_{i+1}}| = s_n + 1$ and $1 \leq i < k$.

Roelants van Baronaigien and Ruskey [4] state the following lemma, which relates the rotation operation with weight sequences.

Lemma 1: [4] Let T be the binary tree with weight sequence $w_1, w_2, \ldots, w_n$. If T' is a binary tree obtained by performing a left rotation on the kth vertex of T, then the weight sequence of T' is

$$w_1, w_2, \ldots, w_{k-1}, w_k + w_{k-w_k}, w_{k+1}, \ldots, w_n.$$

To generate all weight sequences of length n, many algorithms are developed for this purpose [1, 3, 4, 6, 9, 12]. Lucas [2] has shown that the underlying recursion tree of all these algorithms are isomorphic. Roelants van Baronaigien and Ruskey's algorithm [4], referred to as Alg-RR generates not only the binary trees in Gray-code manner, but also the weight sequences in Gray-code order. This helps to ease the discussion in proving the hamiltonian property. Figure 3 shows the recursion tree of Alg-RR. Note that each weight sequence corresponds to a root-to-leaf path in the recursion tree. An n-level

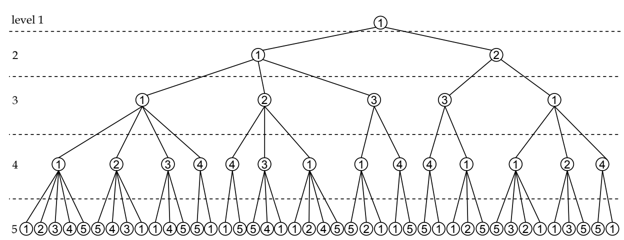

Figure 3: The recursion tree for $n = 5$.

recursion tree is the recursion tree corresponds to weight sequences of length n and can be recursively constructed from the $(n-1)$-level one. As shown in Figure 3, a node x belongs to level k if the distance between x and the root is $k - 1$. In each level k of an n-level recursion tree, we categorize the weight sequences into groups $G_{s_1 s_2 \cdots s_k}$, where

$$G_{s_1 s_2 \cdots s_k} = \{s'_1 s'_2 \cdots s'_n : s'_i = s_i \text{ for } 1 \leq i \leq k\}.$$

A group G_s is said to be a *level-k group* if the length of s is k. The relationship between level-2 groups is explored as follows.

Lemma 2: Let $s = s_1 s_2 \cdots s_n$ be a weight sequence. If $s \in G_{12}$, then there exists a weight sequence $s' = s'_1 s'_2 \cdots s'_n \in G_{11}$ such that $s'_i = s_i$, for $3 \le i \le n$.

Proof: In $T(s)$, vertex s_2 is the right parent of s_1, and vertex s_1 has no child since otherwise 12 is not a prefix of s. The tree $T(s')$ can be obtained by performing a right rotation on the edge (s_1, s_2) in $T(s)$. $\square$

The following lemma, which we called the *rotation lemma*, gives an equivalent description on the adjacency of two weight sequences in RG(n).

Lemma 3: *Rotation Lemma.* Given two weight sequences $s = s_1 s_2 \cdots s_n$ and $s' = s'_1 s'_2 \cdots s'_n$, s and s' are adjacent in RG(n) if and only if s and s' differ in exactly one position k, and $s_1 s_2 \cdots s_k$ and $s'_1 s'_2 \cdots s'_k$ are adjacent in RG(k).

Proof: For necessity, that s and s' differ in exactly one position follows from Lemma 1. Without loss of generality, we assume that $T(s)$ transforms to $T(s')$ by a right rotation on the edge (x, y) with x being the parent of y in $T(s)$. Figure 4 illustrates the structures of $T(s)$ and $T(s')$. Obviously, x corresponds to s_k and s'_k. The k-node trees which correspond to $s_1 s_2 \cdots s_k$ and $s'_1 s'_2 \cdots s'_k$ can be obtained by removing the nodes in subtrees C and D from T and T', respectively. Apparently, the resulting trees can be transformed to each other by a single rotation.

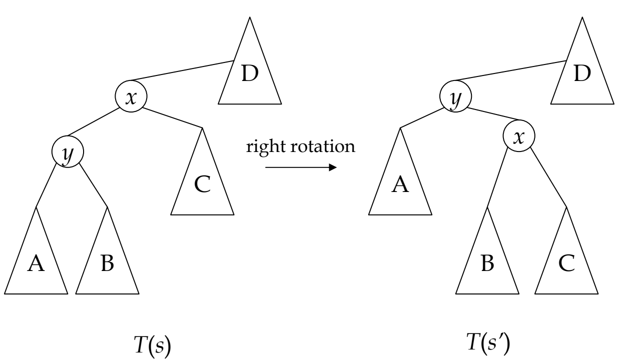

Figure 4: Structures of $T(s)$ and $T(s')$.

For sufficiency, we prove the statement by induction on the difference $n - k$. If $n - k = 0$, the statement holds immediately. Suppose the statement holds when $n - k = i$. When $n - k = i + 1$, consider the two trees obtained by contracting vertices n with their left children from $T(s)$ and $T(s')$. Apparently, the two trees are $T(s_1 s_2 \cdots s_{n-1})$ and $T(s'_1 s'_2 \cdots s'_{n-1})$. According to the induction hypothesis, $T(s_1 s_2 \cdots s_{n-1})$ and $T(s'_1 s'_2 \cdots s'_{n-1})$ are adjacent in RG($n-1$). Recovering the contraction procedure results in that $T(s)$ and $T(s')$ are adjacent in RG(n). By mathematical induction, the statement follows, and the lemma is proven. $\square$

3 The Hamiltonian Cycle in Rotation Graphs of Binary Trees

For $n \le 4$, the existence of Hamiltonian cycles in RG(n) is shown directly by generating the cycles.

Lemma 4: For $n \le 4$, the rotation graph RG(n) contains a Hamiltonian cycle.

Proof: The Hamiltonian cycles in RG(1) and RG(2) are C_1 and C_2, respectively. For $n = 3$, the Hamiltonian cycle in RG(3) is C_5 and is shown below.

$$111 - 112 - 113 - 123 - 121 - 111$$

For $n = 4$, the Hamiltonian cycle in RG(4) is C_{14} and is shown below.

$$1111 - 1112 - 1212 - 1211 - 1231 - 1234 - 1214 - 1114$$

$$-1113 - 1123 - 1124 - 1134 - 1131 - 1121 - 1111$$

$\square$

In the following, we discuss the case where $n \ge 5$. By rotation lemma, $111 \cdots 1$ and $121 \cdots 1$ are adjacent in RG(n). Together with the path generated by tracing Alg-RR, there is a cycle $\hat{C}$, which we called the *base cycle*. The weight sequences in $\hat{C}$ are *covered nodes*, and the rest are *uncovered nodes*. In the following, we show that uncovered nodes can be clustered into cycles which are used to extend $\hat{C}$ so that a Hamiltonian cycle in RG(n) is accomplished.

Definition 1: Given a list $\mathcal{L}$ of n-digit weight sequences generated by Alg-RR, a *stripe* is a segment of weight sequences which belong to an $(n-1)$-level group. Two stripes are *adjacent* if the corresponding weight sequences are continuous in $\mathcal{L}$.

By rotation lemma, we derive the following corollary.

Corollary 5: Given RG(n), the subgraph induced by two adjacent stripes contains a Hamiltonian cycle.

In RG(n), a cycle formed by one stripe or two adjacent stripes is called a *stripe cycle*. By Lemma 2, for every stripe $S \in G_{12}$, there is a corresponding stripe $S' \in G_{11}$ which contains X, where

$$X = \{s' : \text{the (n-2)-digit suffix of } s \text{ and } s' \text{ are the same, where } s \in S \text{ and } s' \in S'\}.$$

We call a cycle C in RG(n) an *extending cycle* if there exist four vertices $x_1, x_2, x_3,$ and x_4, whose induced subgraph contains a C_4, where $C \cap \hat{C} = \phi$, $x_1, x_2 \in C$, and $x_3, x_4 \in \hat{C}$. The four vertices $x_1, x_2, x_3,$ and x_4 are called *boundary vertices* of $C \cup \hat{C}$. For $n = 5$, as shown in Figure 3, only two stripes consist of uncovered nodes. For $n \geq 6$, the stripes which are formed by uncovered nodes can be coupled into cycles if the number of stripes is even. Otherwise, we leave a stripe of size two and couple the others. This can be accomplished because there is at least one pair of adjacent stripes both of which have size two. These cycles are extending cycles according to the following lemma.

Lemma 6: In RG(n), stripe cycles which consist of uncovered nodes are extending cycles.

Proof: Since every uncovered node belongs to G_{12}, it is adjacent to some covered node in G_{11}. If $|S| = 2$, the stripe S consists of the following two weight sequences

$$s = 12a_3a_4 \cdots a_{n-2}(n-1)1, \text{ and}$$

$$s' = 12a_3a_4 \cdots a_{n-2}(n-1)n.$$

The corresponding covered nodes are

$$t = 11a_3a_4 \cdots a_{n-2}(n-1)1, \text{ and}$$

$$t' = 11a_3a_4 \cdots a_{n-2}(n-1)n.$$

Apparently, s, s', t and t' are the boundary vertices of $\hat{C} \cup S$.

For $|S| \geq 3$, the corresponding stripe S' in G_{11} may contain one more weight sequence u since in that case 11 is an edge on the right arm of the binary tree. Let $S = (s, \ldots, t, \ldots v)$, $S' = (s', \ldots, t', \ldots v')$, and the stripe cycle containing S be $C(S)$. The weight sequence u appears either between the pair (s', t') or between (t', v'). Because of the symmetry, we assume that u appears between (t', v'), and s' and t' are adjacent. The boundary vertices of $\hat{C} \cup C(S)$ are s, t, s', and t'. $\square$

By extending $\hat{C}$ with stripe cycles, all uncovered nodes are in a cycle except those which cannot form a stripe. Note that by adding $121\ldots1$ to the uncovered nodes a stripe can be formed. By ALG-RR, these nodes exist only when n is even. Moreover, the number of these nodes is also even because the right arm of $s_1s_2\cdots s_{n-1} = 121\cdots1$ has length $n-1$. Based on the observation given above, a Hamiltonian cycle in RG(n) is constructed as stated in Theorem 7.

Theorem 7: The rotation graph RG(n) contains a Hamiltonian cycle for $n \in \mathbf{N}$.

Proof: By Lemma 4, the theorem holds when $n \leq 4$. For $n \geq 5$, we construct a Hamiltonian cycle in RG(n) as follows. The base cycle is formed by traversing the weight sequences generated by ALG-RR from $111\cdots1$ to $121\cdots1$ and then connect these two weight sequences. If n is odd, the Hamiltonian cycle is accomplished by extending the base cycle with the stripe cycles, and this can be done according to Corollary 5 and Lemma 6. If n is even, besides the stripe cycles, there are uncovered nodes which cannot form a stripe. These nodes belong to the group G_s and are adjacent to the nodes in $G_{s'}$, where $s = 121\cdots1$, $s' = 111\cdots1$, and $|s| = |s'| = n-1$. We list all sequences in G_s and $G_{s'}$ using the recursion tree generated by ALG-RR, as shown in Figure 5. One can see that

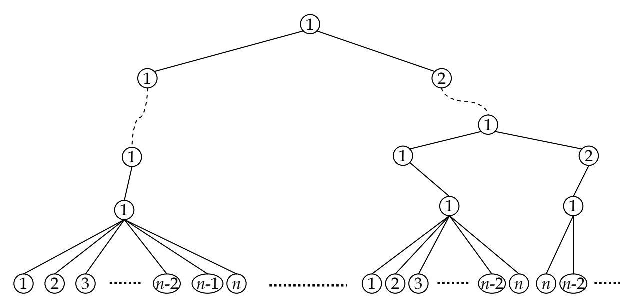

Figure 5: Weight sequences in G_s and $G_{s'}$

for $1 \leq i \leq \frac{n-4}{2}$, weight sequences end with $2i$ and $2i+1$ in G_s and $G_{s'}$ form a C_4. The base cycle is extended by deleting the edge between the weight sequences in $G_{s'}$. The next two weight sequences to $121\cdots11n$ generated by ALG-RR are $121\cdots121n$ and $121\cdots121(n-2)$. Therefore, a C_4 is formed, and a Hamiltonian cycle is accomplished by extending with this C_4. $\square$

4 Concluding Remarks

In this paper, we provide a simpler proof for the existence of a Hamiltonian cycle in rotation graphs. An algorithm for constructing the Hamiltonian cycle can be derived by the proof given above. The corresponding time complexity is $O(\mathcal{C}_n)$, where

$$\mathcal{C}_n = \frac{1}{n+1}\left(\begin{array}{c} 2n \\ n \end{array}\right)$$

is the n^{th} Catalan number. Moreover, the recursion tree structure can be related to *Catalan's triangle*, where entries corresponds to the number of vertices in the recursion tree. Formulae for exact numbers of uncovered vertices, uncovered stripes, and uncovered vertices which do not form a stripe, etc, can all be derived with operands being the entries in Catalan's triangle. Details will be conducted in future works.

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